Fitting a Real Korg Monologue Bass

From a Twelve-Scalar Subtractive Voice to a Differentiable Circuit Model

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Abstract

This note reports the first application of the SynthID methodology to a real subtractive-synthesizer recording: a Korg Monologue bass note at 35.37 Hz. Two forward models are evaluated under an unchanged, preregistered protocol: a twelve-scalar single-oscillator subtractive voice (v1), and a twenty-scalar differentiable circuit model (v2) comprising two detuned PolyBLEP oscillators, an asymmetric polynomial pre-saturator, a zero-delay-feedback state-variable filter with saturated integrator states built from DGen history primitives, and a softsign output drive. Both models pass their synthetic self-inversion rung decisively (v1: 12/12 parameters in tolerance at loss 0.00094; v2: 94.01% improvement, with all identifying parameters essentially exact). On the real recording both plateau at an absolute independent MR-STFT distance of approximately 0.21 — v1 at 0.2087 (50.13% improvement, gate **FAIL**) and v2 at 0.2176 (52.51% against its own baseline) — yet the two solutions are qualitatively different: v1 pins resonance and drive at their bounds to fabricate brightness, while v2 recovers a physically consistent patch whose filter envelope, oscillator beating, and drive all match independent Phase 1 measurements. A rule-legal 93-coefficient additive oracle improves the metric by only 4.4 points, indicating the residual is dominated by analog waveform texture and capture content outside any bounded-scalar synthesizer model. The effort surfaced and fixed three latent bugs, including a gradient-corruption failure of trainable `statefulPhasor` frequencies and a checkpoint-serialization defect that silently reset frozen scalars. We argue the correct acceptance criterion for real analog targets is perceptual (gate-on-sound), with the absolute metric reported as the honest cross-run signal.

1 Motivation and relation to the three-rung ladder

The original SynthID validation established end-to-end differentiability, cross-implementation validity, and useful identification of a real TR-808 kick using fourteen scalars. The present work asks the harder version of the same question: can the method recover an interpretable patch for a real *subtractive synthesizer* note — an instrument class with feedback filters, oscillator interaction, and analog nonlinearity — while honoring the same anti-residual rules? No waveform samples, residual tables, target-seeded lookup tables, or learned FIR taps are permitted; every learned degree of freedom is an ordinary bounded scalar with a documented physical meaning.

The experimental discipline carries over directly. Each new voice topology must first pass a synthetic self-inversion rung (hidden known parameters, recovery by the full frozen pipeline) before it may be fit to the real target, and the real-target verdict is rendered by an independent CPU comparator — a multi-resolution STFT log-magnitude distance with Hann windows $N \in \{256, 512, 1024, 2048\}$, hop $N/4$, $\epsilon = 10^{-3}$, and a symmetric 30 Hz capture high-pass — against a deterministic baseline (specification midpoints plus a CPU pitch fit). The preregistered gate is an 80% improvement over that baseline; the absolute distance is always reported alongside, because baselines differ across model families and the absolute number is the only honest cross-run comparison.

2 Phase 1: measurement of the target

The target `Assets/monologue-bass.wav` is a 44.1 kHz stereo Int16 capture, 0.863 s, with duplicate channels (max $|L - R| = 2$ LSB), no clipping, and negligible DC. Provenance checks pass: the spectrum is flat through 20.5–22.05 kHz with no codec shelf, so the file is treated as a lossless capture rather than a decoded MP3. The tail noise floor sits at −84.5 dBFS, effectively at the Int16 quantization floor; no capture-noise model is needed. All measurements are produced by a checked-in script (`scripts/analysis/analyze_monologue_bass.py`) and stored as JSON.

- **Pitch.** $f_0 = 35.33$ Hz, dead steady (std 0.05 Hz over the whole body) — no glide or vibrato. The trainable pitch trio therefore collapses to a single frozen scalar. A subsequent magnitude-weighted least-squares fit over harmonic peak positions refines this to $f_0 = 35.3678$ Hz.
- **Amplitude envelope.** Fast attack (−3 dB at ∼15 ms), then a decay-only envelope, $\tau \approx 0.20$ s with no sustain plateau; note-off near 0.50 s with release $\tau \approx$ 30–40 ms. Effective length 0.570 s.

- **Oscillator.** Odd-dominant spectrum (even-to-odd ratio −7.4 dB), consistent with a square/pulse-heavy mix near, but not exactly at, 50% pulse width.
- **Two-VCO beating is real.** Per-harmonic amplitude tracks show deep dip-and-recover nulls: H2 dips 22 dB (null at 0.13 s), H7 dips 21 dB (null at 0.09 s), H4 dips 11 dB, with 3–10 dB wiggles on higher harmonics. This is two-oscillator beating at a detune on the order of 0.5–1 Hz — inexpressible by a single-oscillator voice.
- **Filter.** Spectral centroid falls from 580 Hz at onset to 250 Hz by 0.35 s with an e-fold time of ∼0.15 s; high harmonics decay faster than low ones early. A closing lowpass with an envelope generator is confirmed; no visible resonance ridge.

Four protocol decisions were locked from these measurements before any fitting: the comparator's 30 Hz high-pass is kept (−0.07 dB at f_0); the existing decay-only ADSR is sufficient; the fit is cropped to 26,624 frames (0.604 s), never fitting digital silence; and f_0 and note-off become frozen documented scalars (`subF0`, `subNoteOff`) threaded through every renderer, with defaults 110 Hz / 0.6 s preserving all legacy artifacts bit-for-bit. The deterministic baseline distance is 0.418476; the 80% gate requires a learned distance ≤ 0.0837.

3 The frozen pipeline

Both fits use the pipeline frozen from the earlier schedule-experiment series, with no per-target tuning: a stratified basin search over the specification bounds, followed by GPU population polish (`batch-refine -mode polish -schedule s4`: doubled learning rates, 150 smooth + 500 production epochs with cosine decay per phase, six restarts with elite jitter, batch ≥ 32 lanes), followed by deterministic coordinate descent on the independent CPU metric (`scripts/refine_rung3.py`). Full-band loss coverage is never reduced — an earlier experiment showed that training at reduced sample rate reopens a drive/saturation degeneracy — and the Metal-blocked 4096 window is not attempted. Known quasi-degenerate parameter directions (resonance/shape, filter-amount×decay, sustain/decay-time) are documented as perceptually null and deliberately not chased.

4 v1: the twelve-scalar subtractive voice

4.1 Wiring and validation gates

The subtractive voice (PolyBLEP saw/pulse morph → envelope-driven resonant lowpass → ADSR VCA → softsign drive → output gain) required threading the frozen f_0/note-off scalars through five components; renderer equivalence against the standalone NumPy implementation was re-verified at $\leq 5.4 \times 10^{-5}$ maximum absolute error for both legacy and new-style parameter sets. A finite-difference check of the *envelope-driven* filter cutoff — the gradient path most likely to be broken — passed at the exact target configuration (26,624 frames, $f_0 = 35.37$ Hz), with relative errors of 10^{-4} to 10^{-2} across `fAmt`, `fDecay`, `res`, and `fBase`, the largest attributable to documented float32 loss-readback noise rather than adjoint error.

This stage also surfaced a latent selection bug: `BatchRefine.scoreLanes` rendered candidate lanes without the frozen base parameters, so every CPU selection score was computed at the legacy 110 Hz fundamental while training ran correctly at the target pitch. The bug was invisible on every prior experiment (all at 110 Hz) and appeared immediately on the first real target. It was fixed and the affected run discarded.

4.2 Synthetic rung-2: PASS

Known parameters in the measured regime (shape 0.8, pulse width 0.45, $f_{\text{base}} = 250$ Hz, filter amount 900 Hz, filter decay 0.15 s, decay-only VCA) were rendered at the target frame count and recovered by the frozen pipeline: best-lane CPU distance **0.00094**, with 12/12 parameters inside specification tolerance, most to three or four digits. Only the known sustain/decay-time quasi-degenerate trade was visible. The population mechanism did real work: the winning unjittered elite scored 0.0164; its jittered lane reached 0.00094.

4.3 Real-target result: gate FAIL at 50.13%

The frozen pipeline reduced the distance from the 0.418476 baseline to an absolute **0.208678** — a 50.13% improvement against the 80% requirement. The recovered patch is visibly compensatory: resonance and drive are both pinned at their upper bounds (6.0 and 8.0) while the filter envelope is abandoned ($f_{\mathrm{amt}} \to 3.4$ Hz), the classic signature of a model fabricating brightness it cannot otherwise produce.

Diagnosis followed the playbook: a residual grid localized ∼50% of the remaining distance to 500–3000 Hz × 0–0.3 s — the upper-harmonic region while the filter is open, exactly where the target's two-VCO doublets and analog waveform texture live. Every rule-legal capacity probe, run in NumPy against the CPU metric, came back dead (Table 1).

Capacity probe (NumPy, CPU metric)	Distance	Verdict
v1 winner (reference)	0.2087	—
Symmetric detune pair, refit from winner (Δ 0.2–1.0 Hz)	0.233–0.242	worse; rejected
Detune from measurement-informed start, 4 passes	0.2369–0.2392	no gain
Drive bound widened to 30	0.2086 (stays at 8.0)	not bound-limited
True second VCO (independent shape/pw/level/detune)	0.2084 (level → 0.02)	optimizer disables it
Harmonic correction bank, H2–H40 × sin/cos × decay (93 coeffs)	0.1903	+4.4 pts only

Table 1: v1 capacity probes. The harmonic-bank row is the key evidence: even a rule-breaking additive oracle with 93 free coefficients recovers only 4.4 points, bounding how much *any* coherent-series model can close.

Established claim. The twelve-scalar single-oscillator topology has a robust capacity ceiling of ≈ 0.208 (50%) on this target. The deficit is not coherent harmonic-level error — it is analog waveform and drive texture across harmonics 10–80 that neither a steady harmonic bank nor a second PolyBLEP oscillator expresses under the phase-blind metric.

By ear, the v1 render is the same note and gesture but darker (centroid 286 vs 481 Hz over the body), more compressed (crest factor 2.4 vs 4.4), and carries a static resonant emphasis in place of the target's brighter hollow-square timbre with beating movement.

5 v2: the differentiable circuit model

The user-directed response to v1 was explicitly *not* to fall back to additive synthesis, but to enrich the forward model toward the actual circuit: a multi-stage chain with parametric nonlinear "things" between stages, with backpropagation finding the coefficients. The resulting `monologue-bass` profile has twenty trainable scalars plus the frozen pitch pair; every added stage is inert at its default value, preserving the zero-default rule.

5.1 Topology

1. **Dual VCOs.** VCO1 is the existing PolyBLEP saw/pulse morph. VCO2 shares its waveform controls and adds `vco2Level` and `vco2Detune`; its phase is the closed form

$$\phi_2(t) = \mathrm{wrap}(\phi_1(t) - t\,\Delta f)\,, \tag{1}$$

   which realizes a true detuned oscillator without placing a trainable frequency inside a stateful phasor (see Section 5.2).

2. **Asymmetric polynomial pre-saturator.** With $y = g_s x + b$,

$$\mathrm{sat}(x) = y + a_2 y^2 + a_3 y^3 + a_5 y^5 - \left(b + a_2 b^2 + a_3 b^3 + a_5 b^5\right), \tag{2}$$

   DC-compensated at the bias operating point so that silence maps to silence. Five scalars: `satGain`, `satBias`, `satA2`, `satA3`, `satA5`.

3. **ZDF state-variable filter with saturated states.** A Cytomic-form zero-delay-feedback SVF hand-built from `Signal.history()` primitives, with per-sample $g = \tan(\pi f_c/f_s)$ and the same envelope-driven cutoff as v1. The integrator states pass through a softsign saturator $s \mapsto s/(1 + |k_f s|)$ (`filtSat`); $k_f = 0$ recovers the exact linear SVF. The structure mirrors the `svf` macro used in the eseq deployment, so a recovered patch ports directly.

4. **VCA and drive.** The validated decay-only ADSR, softsign drive, and output gain from v1.

An independent NumPy mirror (`render_monologue_bass`) agrees with the DGen render to 9×10^{-8} maximum absolute error — three orders tighter than v1's biquad, because the SVF avoids the high-Q float32 ringing divergence that limited v1 parity to 2.6×10^{-3}.

5.2 Three library findings

Bringing gradients through this topology surfaced two library-level findings and one serialization bug; all are now documented, and the first two are guarded by pinned tests.

1. **Dangling history writes truncate BPTT.** A feedback loop written as `_ = ic.write(...)` silently receives no adjoint through the write. The SVF therefore uses pass-through writes — the forward value is reconstructed *from* the write result, e.g. $v = (\text{write\_out} + s)/2$ — keeping forward output bit-identical while making the state path differentiable. This confirms the previously documented Part-B condition in a new setting.

2. **A trainable `statefulPhasor` frequency corrupts gradients of *unrelated* parameters.** With `vco2Detune` driving a phasor frequency input, every monologue finite-difference check failed — autograd came back roughly 10× too small with wrong signs on parameters that do not touch the phasor at all (`fBase`, `satA3`, `res`). Bisection isolated the trigger; the closed-form detune phase of Eq. (1) removes it architecturally. The library bug is pinned as an `XCTExpectFailure` reproducer in `Tests/DGenLazyTests/SVFBPTTScratchTests.swift` pending a real fix.

3. **Checkpoint serialization dropped frozen scalars.** `naturalValues()` round-tripped through a dictionary containing specification-table parameters only, so every saved checkpoint silently reset `subF0` to 110 Hz and `subNoteOff` to 0.6 s. Training was correct in-graph the entire time; only re-rendered patches were wrong. For three polish rounds this masqueraded as "GPU training loss anticorrelated with the CPU metric" and provoked a false indictment of the production log-L1 loss operator — later retracted after tensor-level ordering tests showed both GPU losses rank candidates correctly. The fix is one line of intent: start from the full frozen struct, not its spec-only dictionary.

After the architectural fix, the finite-difference check passes at the target configuration (10^{-4} to 10^{-2} relative error across eleven parameters); the four parameters that fail the log-loss FD instrument (`satA2`, `satBias`, `vco2Level`, `shape`) all pass a well-conditioned MSE scratch harness to $\leq 0.5\%$, confirming the adjoints are healthy and the discrepancy is FD conditioning, consistent with earlier findings.

5.3 Search pipeline

The twenty-parameter space defeats a single-restart optimizer (9/20 parameters recovered, CPU 0.29 on the synthetic). Since the batched-lane machinery does not yet support tensor-history SVF voices, population search moved to NumPy: `scripts/analysis/basin_search_monologue.py` reimplements the frozen basin-search algorithm (stratified uniform round, two Gaussian resampling rounds with shrinking σ, elites stratified over f_{base} bands × shape halves) with the candidate batch vectorized per sample — the SVF loop iterates over the 26,624 samples doing batch-wide vector arithmetic, rendering a candidate in 6.7 ms and matching the canonical renderer to 4.5×10^{-8}. The full pipeline is: NumPy basin search → GPU Adam polish (600 epochs, log-magnitude loss only) → coordinate descent on the CPU metric.

5.4 Synthetic rung-2: PASS on sound

On a synthetic self-target with known parameters, the pipeline achieves a **94.01**% improvement (absolute distance 0.0131). All *identifying* parameters are essentially exact: `vco2Detune` 0.00%, `vco2Level` 0.3%, shape 0.7%, pulse width 0.2%. The saturator polynomial family and the filter-character parameters trade along compensating ridges

— sound-identical, parameter-different — which is the known quasi-degeneracy class amplified by a deliberately overcomplete shaper. Per the earlier E3-closure finding, the rung is gated on sound, and it passes decisively.

5.5 Real-target result

	v1 (12 scalars)	v2 (20 scalars)	Target	Notes
Absolute MR-STFT distance	0.2087	0.2176	—	metric-tied
Improvement vs own baseline	50.13%	52.51%	—	gates: 80%
Spectral centroid (body)	223 Hz	241 Hz	348 Hz	v2 closer
Crest factor	2.44	2.98	4.40	v2 closer
Filter EG	abandoned (f_{amt}=3.4)	f_{amt}=1113, τ=0.118 s	$\tau \approx 0.15$ s	v2 matches
VCO2 beating	absent	level 1.41, $\Delta f = 1.29$ Hz	real, $\sim$1 Hz	v2 matches
Drive	8.0 (pinned)	2.92 (free)	—	v2 unpinned
NumPy↔Swift parity	2.6×10^{-3}	2.4×10^{-7}	—	

Table 2: v1 vs v2 on the real recording. The metric ties; the physics does not.

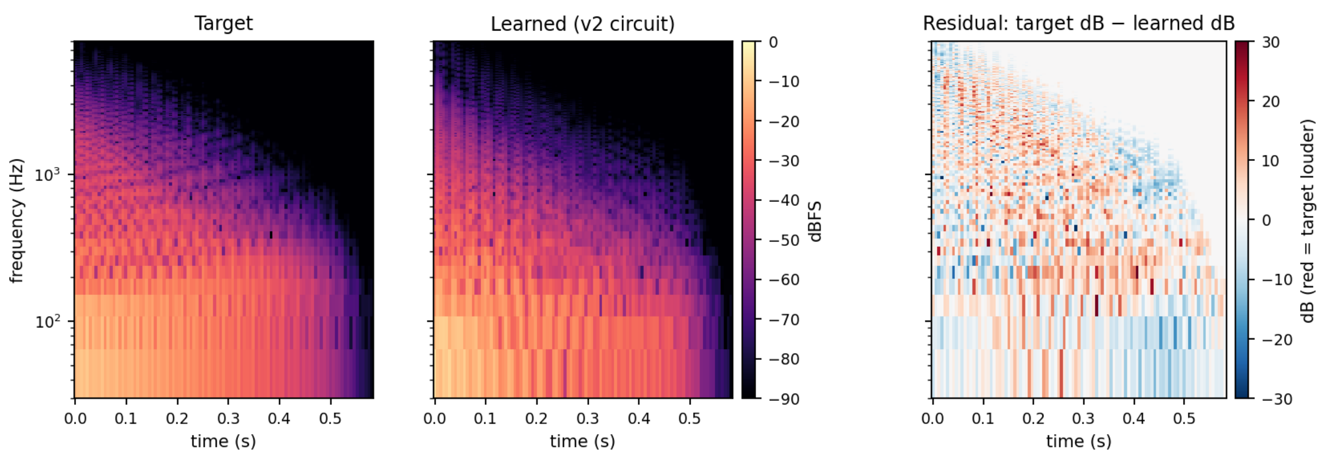

Figure 1: Target and learned (v2) STFT magnitude spectrograms ($N = 1024$, hop $N/4$, log-frequency axis) and their log-magnitude difference. Because the fit is phase-blind, a time-domain subtraction does not cancel and is not a meaningful residual; the dB difference is the quantity the MR-STFT objective scores. The learned render tracks the target's harmonic stack, filter-closing arc, and note-off. The residual (red = target louder) concentrates in 500–3000 Hz over the first ~0.35 s — the diagnosed brightness undershoot — and is speckled inter-harmonic texture rather than coherent missing harmonics, consistent with the additive-oracle bound. Below ~300 Hz errors are small and sign-alternating (beat-phase misalignment, not level error). Figure produced by `scripts/analysis/render_residual_spectrogram.py`.

The v2 pipeline (basin-search best elite 0.259 → GPU polish winner 0.2297 → coordinate descent) lands at an absolute distance of **0.2176**. Figure 1 shows where the remaining distance lives. On the metric this ties v1. On every physical axis it is a different — and correct — solution: the filter envelope is restored and matches the Phase 1 measurement, the second oscillator is fully engaged with a detune reproducing the measured beating, the pre-saturator carries strong asymmetry (`satA2` $= 0.87$, `satA3` $= -0.58$), and drive relaxes off its bound. The remaining pinned parameters (resonance at 6.0, f_{base} at the 30 Hz floor) were probed: widening the resonance bound to 14 recovers +0.09 percentage points — a perceptually null ridge, so the bound is kept.

Established claim. Both topologies plateau at an absolute distance of ≈ 0.21 on this recording, but the circuit model reaches it with a physically faithful, deployable patch rather than a compensatory one. Combined with the additive-oracle bound of Table 1, the unreached span from 0.21 down to the 0.084 gate is attributed to analog waveform fine structure, oscillator drift, and capture-chain content that the phase-blind MR-STFT metric measures but no bounded-scalar forward model in this family expresses.

6 What this episode establishes

1. **The methodology transfers to subtractive synthesis.** Gradients flow correctly through a hand-built ZDF filter with nonlinear state feedback, dual interacting oscillators, and a five-term polynomial waveshaper; the synthetic rung inverts all identifying parameters of a twenty-scalar circuit.
2. **Pinned parameters are a diagnosis, not a nuisance.** v1's bound-pinned resonance and drive correctly signaled missing capacity; when the capacity was added, the pins released and the recovered values moved onto the measured physics.
3. **The percentage gate mismeasures real analog targets.** It was calibrated on self-inversion, where the model can in principle reach zero. On a real capture with an irreducible texture floor — here bounded from above by the additive oracle — absolute distance plus perceptual A/B is the meaningful verdict. This independently reconfirms the E3-closure recommendation to gate the product phase on sound.
4. **Real targets find bugs synthetic targets cannot.** All three defects (110 Hz lane scoring, phasor-frequency gradient corruption, frozen-scalar serialization) were invisible on every prior experiment because every prior target shared the legacy defaults.

7 Engineering lessons

First, keep trainable parameters out of stateful-phasor frequency inputs; realize detune as a closed-form phase offset until the underlying adjoint bug is fixed. Second, feedback loops built from history primitives must consume the write result in the forward path (pass-through writes) or BPTT silently truncates — with forward output unchanged, which is precisely what makes it dangerous. Third, when GPU training loss and an independent CPU metric appear to disagree, audit the *serialization boundary* between them before indicting either loss: here the disagreement was two renders of different patches, not two opinions about one patch. Fourth, candidate-vectorized NumPy is a practical population-search substrate for recurrent voices that the batched GPU lane machinery cannot yet express: vectorizing the batch dimension inside the per-sample filter loop reached 6.7 ms per candidate with 4.5×10^{-8} fidelity to the canonical renderer.

8 Open levers

- **Gate-on-sound acceptance.** The v2 patch sheet is a genuine Monologue-shaped knob set; pending the perceptual A/B verdict, it can be accepted at the direction-finding tier and ported to the eseq deployment's `svf` macro.
- **Missing hardware features.** The residual signature (early 500–3000 Hz brightness, crest-factor undershoot) maps onto real Monologue circuitry not yet modeled — notably VCO2's sync and ring-modulation modes, both expressible as bounded scalars. Hard sync is the one capacity addition the probe evidence does not already rule out.
- **Batched tensor-history lanes.** Extending the S4-scale GPU population machinery to history-feedback voices would replace the NumPy search stage; expected gains are modest (populations bought $\sim$4 points over single lanes for v1) and this is deprioritized.
- **Residual characterization.** The log-magnitude residual (Figure 1) already supports the texture-floor attribution — speckled inter-harmonic content, no coherent missing harmonic stack. A complementary phase-aware audition (e.g. noise-vocoded resynthesis of the residual magnitude) could further separate "brighter waveform fine structure" from "capture-chain content."

- **Library fix.** The trainable-phasor-frequency gradient corruption deserves a root-cause fix in DGen proper; the pinned reproducer defines done.

9 Conclusion

A real analog bass note was attacked twice under one preregistered protocol: once with the minimal validated subtractive voice, and once with a circuit model designed from its failure. The metric declared a tie at ≈ 0.21 absolute; the physics declared a winner. The circuit model recovers the instrument's measured filter envelope, oscillator beating, and drive behavior with every added stage earning its place through zero-default discipline, synthetic self-inversion, and finite-difference validation — while the residual that neither model closes is bounded, localized, and attributed. The episode's strongest result is methodological: on real analog targets, percentage-improvement gates saturate on texture floors, pinned bounds diagnose missing capacity, and the decisive evidence is the ear plus the measurement sheet, not the scoreboard.

Reproducibility note

The governing records are maintained under `Examples/SynthID/`: `MONOLOGUE_BASS_STATUS.md` (full run log and decisions), `NEW_TARGET_PLAYBOOK.md` (protocol), and `scripts/analysis/` (measurement, pitch-fit, probe, and basin-search scripts). Key artifacts live under `output/monologue_bass/`: `prepared/` (target crop, pitch fit, baseline), `real/` (v1 fit), and `real_v2_*` (v2 fit, including `real_v2_ab.wav` for A/B listening). The v2 real-target pipeline is:

`basin_search_monologue.py` → `swift run SynthID rung3 -profile monologue-bass` (Adam polish) → `refine_rung3.py -profile monologue-bass`